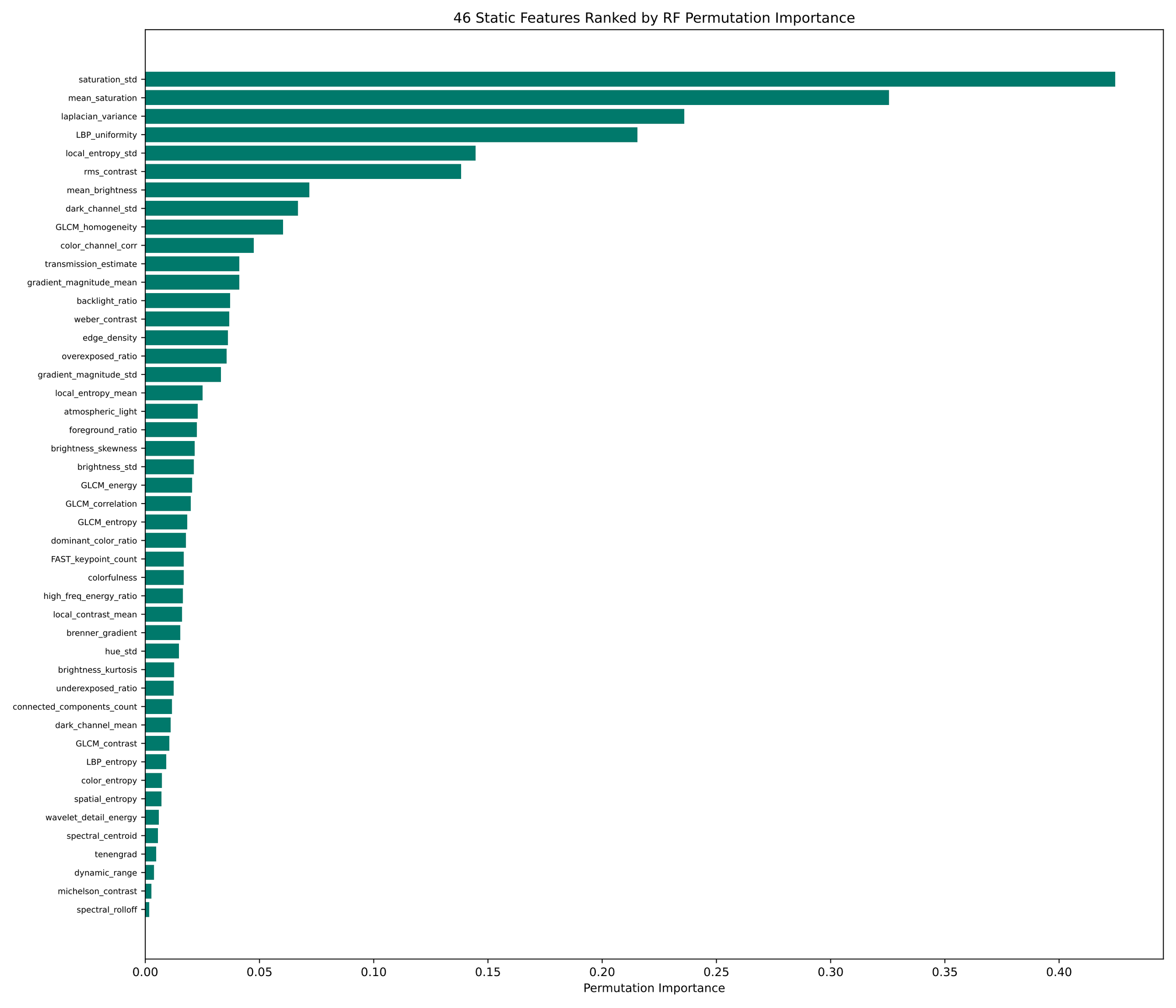
All 56 Features Ranked by Spearman Correlation with Recall Gap

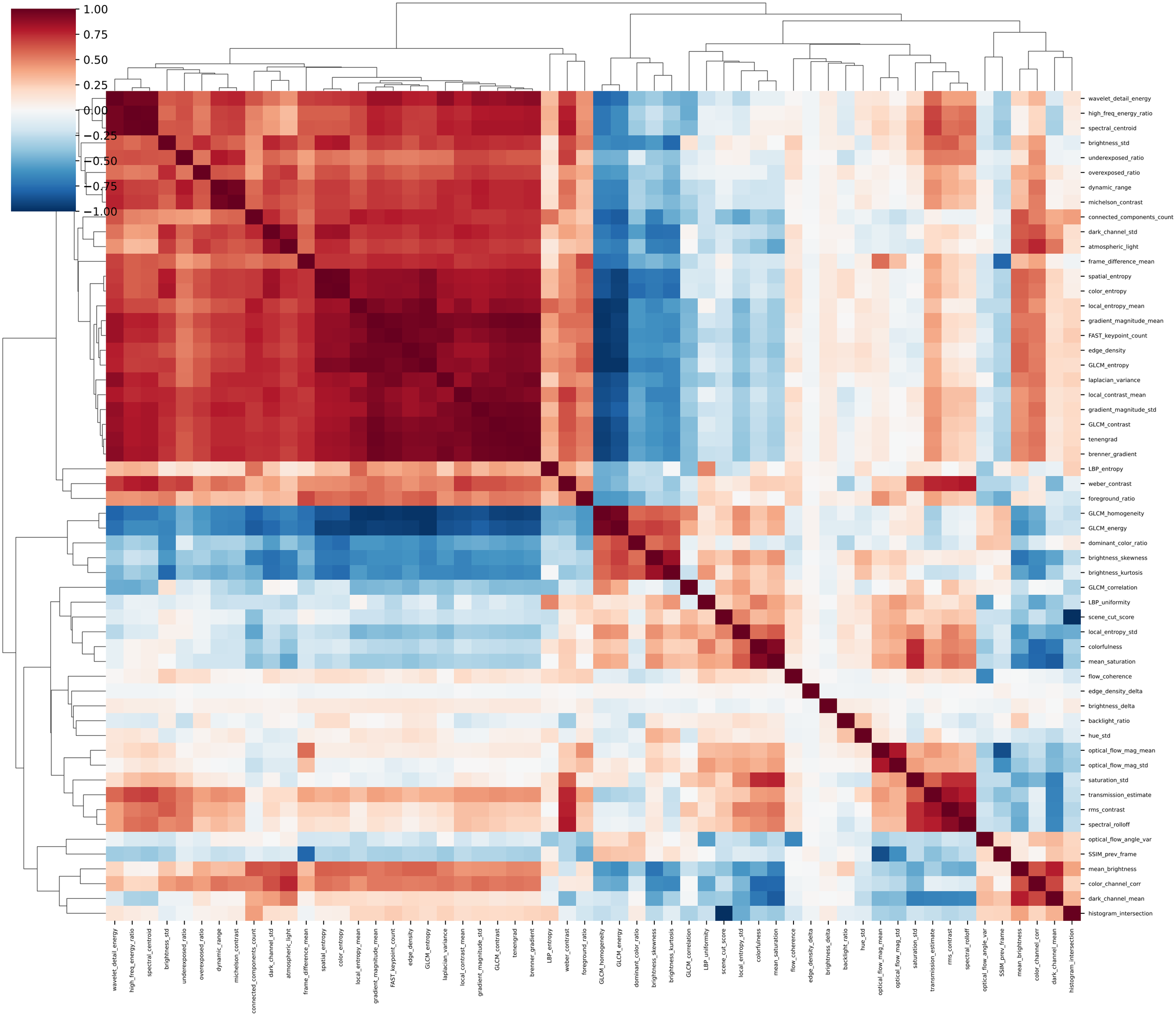
rms_contrast
transmission_estimate
weber_contrast
spectral_rolloff
spectral_centroid
high_freq_energy_ratio
saturation_std
dark_channel_mean
brightness_std
underexposed_ratio
wavelet_detail_energy
brenner_gradient
optical_flow_mag_mean
mean_saturation
GLCM_contrast
SSIM_prev_frame
gradient_magnitude_std
tenengrad
frame_difference_mean
dynamic_range
local_contrast_mean
FAST_keypoint_count
gradient_magnitude_mean
foreground_ratio
michelson_contrast
laplacian_variance
local_entropy_std
colorfulness
optical_flow_mag_std
edge_density
GLCM_entropy
color_entropy
GLCM_homogeneity
spatial_entropy
mean_brightness
dominant_color_ratio
local_entropy_mean
overexposed_ratio
GLCM_energy
optical_flow_angle_var
brightness_kurtosis
dark_channel_std
connected_components_count
flow_coherence
color_channel_corr
LBP_uniformity
backlight_ratio
atmospheric_light
brightness_skewness
scene_cut_score
histogram_intersection
hue_std
GLCM_correlation
edge_density_delta
brightness_delta
LBP_entropy

−0.3 −0.2 −0.1 0.0 0.1 0.2 0.3 0.4 0.5

Spearman rho with recall_gap


46 Static Features Ranked by RF Permutation Importance
saturation_std
mean_saturation
laplacian_variance
LBP_uniformity
local_entropy_std
rms_contrast
mean_brightness
dark_channel_std
GLCM_homogeneity
color_channel_corr
transmission_estimate
gradient_magnitude_mean
backlight_ratio
weber_contrast
edge_density
overexposed_ratio
gradient_magnitude_std
local_entropy_mean
atmospheric_light
foreground_ratio
brightness_skewness
brightness_std
GLCM_energy
GLCM_correlation
GLCM_entropy
dominant_color_ratio
FAST_keypoint_count
colorfulness
high_freq_energy_ratio
local_contrast_mean
brenner_gradient
hue_std
brightness_kurtosis
underexposed_ratio
connected_components_count
dark_channel_mean
GLCM_contrast
LBP_entropy
color_entropy
spatial_entropy
wavelet_detail_energy
spectral_centroid
tenengrad
dynamic_range
michelson_contrast
spectral_rolloff
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
Permutation Importance

Pairwise Feature Correlations (Clustered)



Top 6 Features vs Recall Gap

rms_contrast
rho=0.475

transmission_estimate
rho=0.438

weber_contrast
rho=0.430

spectral_rolloff
rho=0.404

spectral_centroid
rho=0.368

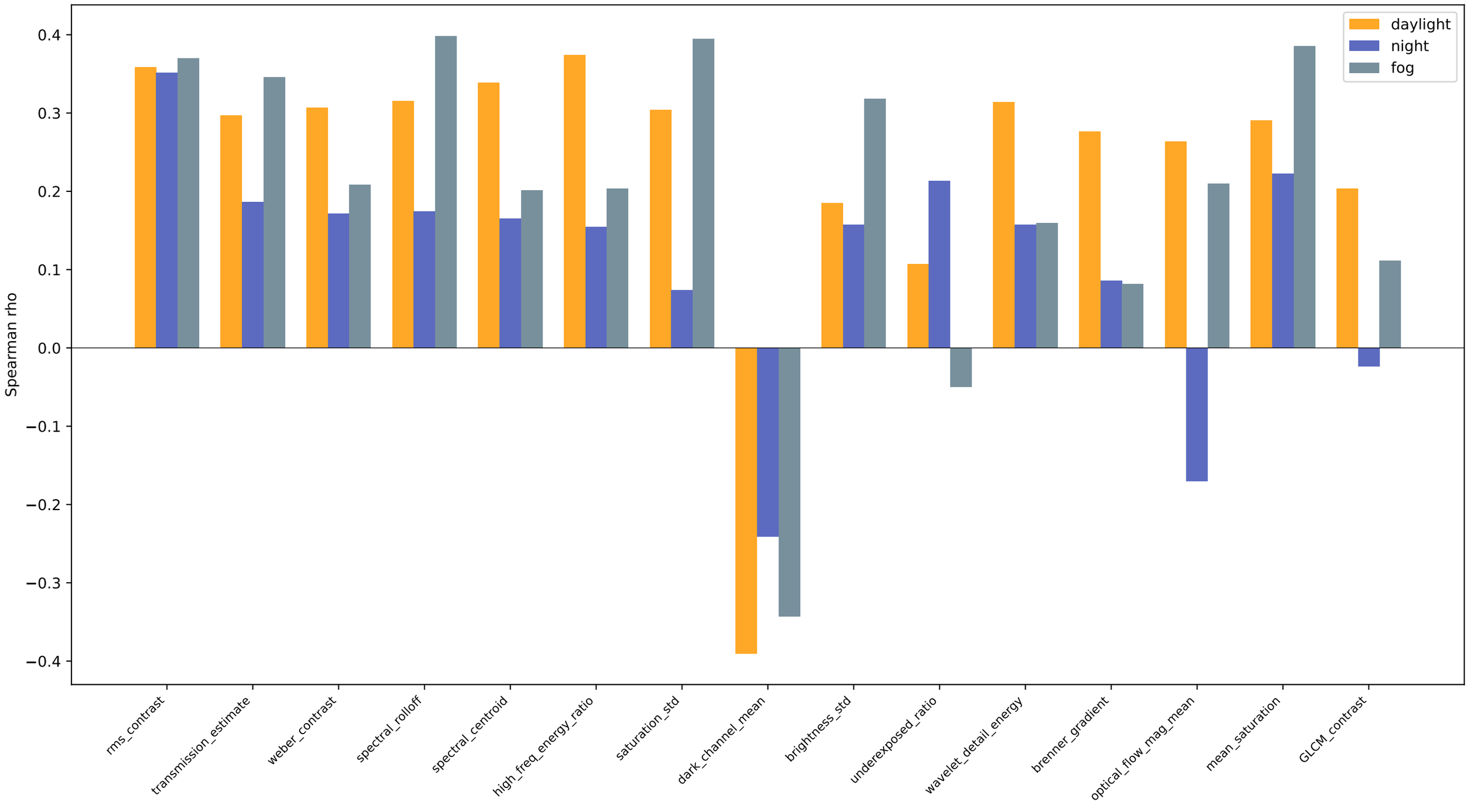
high_freq_energy_ratio
rho=0.356

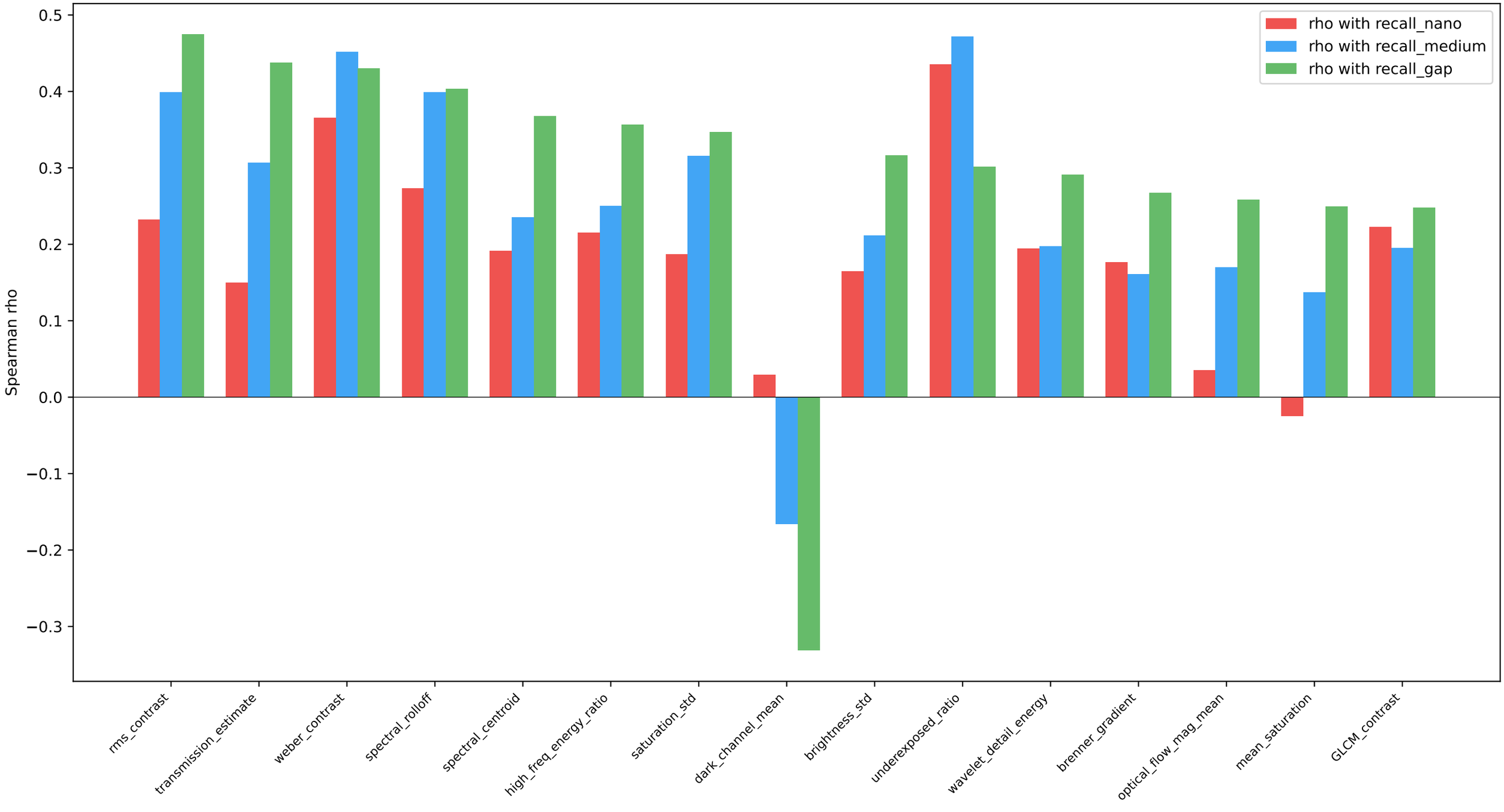
recall_gap

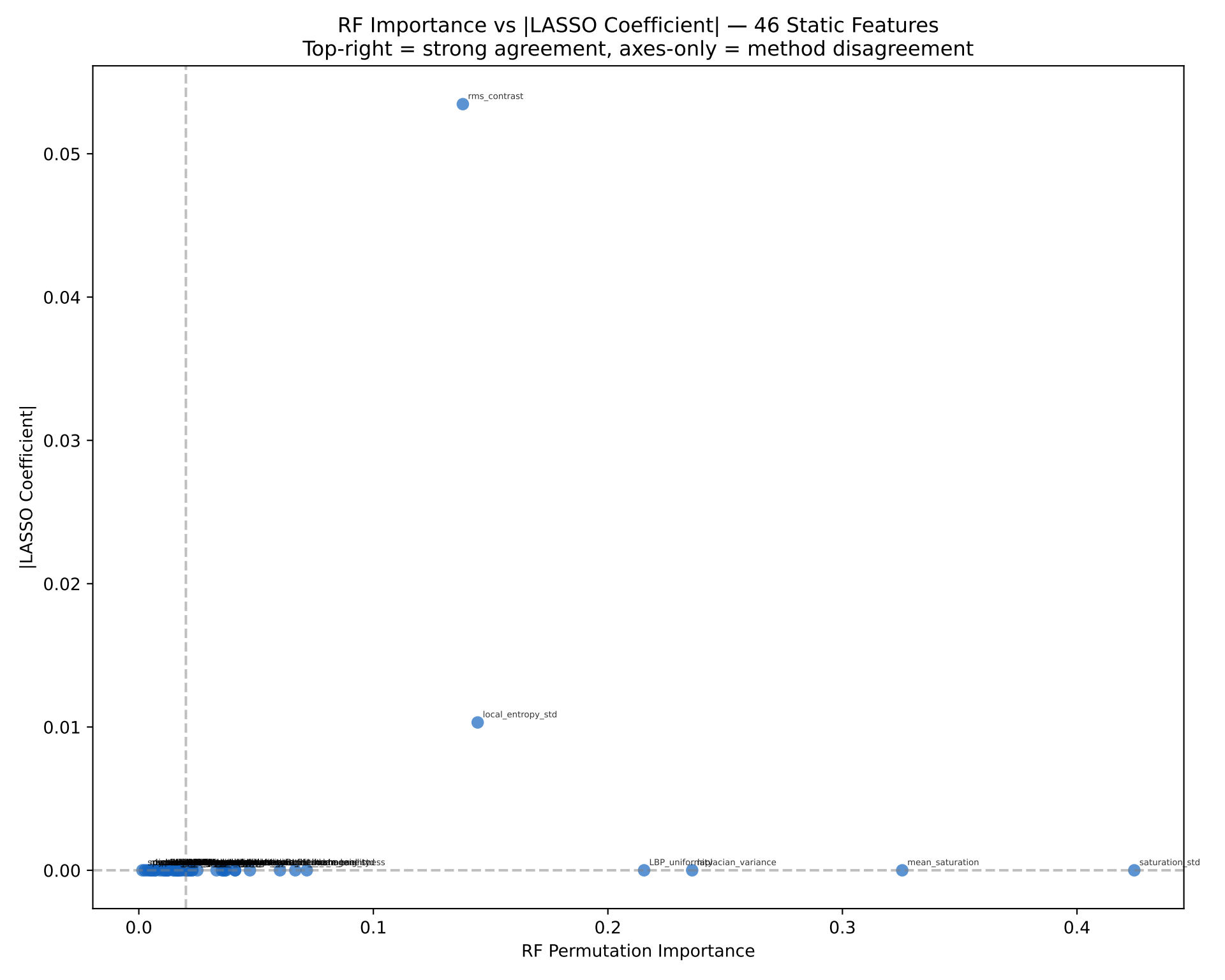
spectral_rolloff

spectral_centroid

high_freq_energy_ratio


Spearman Correlation by Weather Condition (Top 15 Features)
daylight
night
fog
Spearman rho
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
rms_contrast
transmission_estimate
weber_contrast
spectral_rolloff
spectral_centroid
high_freq_energy_ratio
saturation_std
dark_channel_mean
brightness_std
underexposed_ratio
wavelet_detail_energy
brenner_gradient
optical_flow_mag_mean
mean_saturation
GLCM_contrast


Independent Model Correlations — Top 15 Features
* = |rho_nano| >> |rho_medium| (most valuable for switching)
rho with recall_nano
rho with recall_medium
rho with recall_gap
Spearman rho
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
rms_contrast
transmission_estimate
weber_contrast
spectral_rolloff
spectral_centroid
high_freq_energy_ratio
saturation_std
dark_channel_mean
brightness_std
underexposed_ratio
wavelet_detail_energy
brenner_gradient
optical_flow_mag_mean
mean_saturation
GLCM_contrast


RF Importance vs |LASSO Coefficient| — 46 Static Features
Top-right = strong agreement, axes-only = method disagreement
rms_contrast
local_entropy_std
LBP_uniformity
Laplacian_variance
mean_saturation
saturation_std
|LASSO Coefficient|
RF Permutation Importance
0.00
0.01
0.02
0.03
0.04
0.05
0.0
0.1
0.2
0.3
0.4

Recommended Feature Subset (26 features: 16 static + 10 temporal)
Static: RF importance > median AND not redundant | Temporal: kept unconditionally

Feature	Category	Spearman rho	RF Importance	LASSO Coef	Keep Reason
mean_brightness	illumination	-0.1684	0.0716	-0.0000	RF > median, not redundant (LASSO disagrees)
brightness_std	illumination	0.3164	0.0213	0.0000	RF > median, not redundant (LASSO disagrees)
brightness_skewness	illumination	-0.0385	0.0216	-0.0000	RF > median, not redundant (LASSO disagrees)
overexposed_ratio	illumination	0.1569	0.0354	-0.0000	RF > median, not redundant (LASSO disagrees)
backlight_ratio	illumination	-0.0596	0.0371	-0.0000	RF > median, not redundant (LASSO disagrees)
weber_contrast	contrast	0.4303	0.0367	0.0000	RF > median, not redundant (LASSO disagrees)
rms_contrast	contrast	0.4746	0.1382	0.0535	RF > median, not redundant, LASSO agrees
dark_channel_std	haze	0.1409	0.0667	-0.0000	RF > median, not redundant (LASSO disagrees)
transmission_estimate	haze	0.4379	0.0411	0.0000	RF > median, not redundant (LASSO disagrees)
laplacian_variance	sharpness	0.2136	0.2359	0.0000	RF > median, not redundant (LASSO disagrees)
LBP_uniformity	texture	0.0599	0.2154	-0.0000	RF > median, not redundant (LASSO disagrees)
local_entropy_std	entropy	0.2130	0.1445	0.0103	RF > median, not redundant, LASSO agrees
mean_saturation	color	0.2495	0.3255	0.0000	RF > median, not redundant (LASSO disagrees)
saturation_std	color	0.3469	0.4244	0.0000	RF > median, not redundant (LASSO disagrees)
color_channel_corr	color	-0.0696	0.0474	0.0000	RF > median, not redundant (LASSO disagrees)
foreground_ratio	spatial	0.2185	0.0225	0.0000	RF > median, not redundant (LASSO disagrees)
optical_flow_mag_mean	temporal	0.2586	N/A	N/A	temporal (kept unconditionally)
optical_flow_mag_std	temporal	0.2053	N/A	N/A	temporal (kept unconditionally)
optical_flow_angle_var	temporal	-0.1446	N/A	N/A	temporal (kept unconditionally)
frame_difference_mean	temporal	0.2339	N/A	N/A	temporal (kept unconditionally)
SSIM_prev_frame	temporal	-0.2472	N/A	N/A	temporal (kept unconditionally)
histogram_intersection	temporal	-0.0249	N/A	N/A	temporal (kept unconditionally)
brightness_delta	temporal	0.0022	N/A	N/A	temporal (kept unconditionally)
edge_density_delta	temporal	-0.0048	N/A	N/A	temporal (kept unconditionally)
flow_coherence	temporal	0.0703	N/A	N/A	temporal (kept unconditionally)
scene_cut_score	temporal	0.0249	N/A	N/A	temporal (kept unconditionally)